

## 2.4 Time Series Classification with Pixel-Set Encoders and Temporal Self-Attention<sup>1</sup>

The PseTae (Pixel-Set Time-series Attention Encoder) architecture is a deep learning model optimized for satellite imagery classification, designed to improve both speed and accuracy for tasks like agricultural monitoring. PseTae treats each image patch as an unordered set of pixels, focusing directly on the spectral information without the constraints of a strict spatial structure. Furthermore, it replaces computationally intensive, sequential recurrent networks with a self-attention mechanism that analyzes the entire time series in parallel, dynamically weighing the importance of different seasonal patterns [1].

To evaluate the performance of the PseTae architecture, the experimental dataset was standardized using a consistent baseline of 5x5-pixel patches observed over an eight-month temporal window (March–October). Within this framework, three distinct input configurations were tested to assess feature sensitivity: a spectral-only case, a vegetation-index-only case, and a combined case integrating both.

To ensure a consistent classification pipeline, the raw satellite imagery (for training and testing) was standardized into time-series-compatible tensors of shape  $(T = 8, C_n, S = 25)$ , where  $T$  represents the temporal sequence and  $S$  denotes the flattened spatial patch. The spectral dimension  $C_n$  was dynamically configured based on the input modality, ranging from 5 to 15 channels depending on whether vegetation indices, raw spectral bands, or both were used.

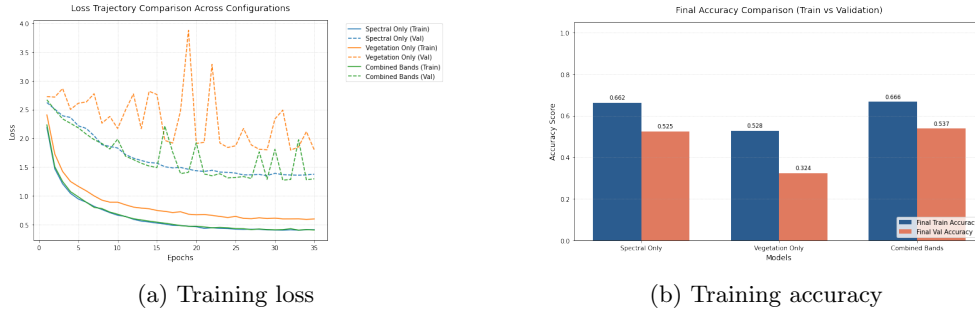
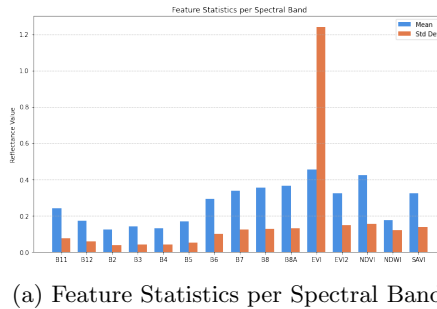


Figure 7: 5x5 — 8 months — Band combinations — training

While the inclusion of spectral bands leads to a smooth transformer training, the inclusion of Vegetation Indices (VIs) introduced numerical instability due to significant discrepancies in feature-wise variance. Because indices like the EVI exhibit distributions and scales that differ from those of raw spectral reflectance bands, their inclusion created an imbalance in gradient flow during backpropagation. This mismatch effectively conditioned the loss landscape, making the model overly sensitive to high-variance features.



(a) Feature Statistics per Spectral Band

Figure 8: Bands values

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Consequently, the optimization process encountered frequent 'spikes' in loss (fig. 7a), preventing stable convergence and leading the model to default to predicting a single class as a local optimum, failing to learn meaningful discriminative features (fig. 9b).

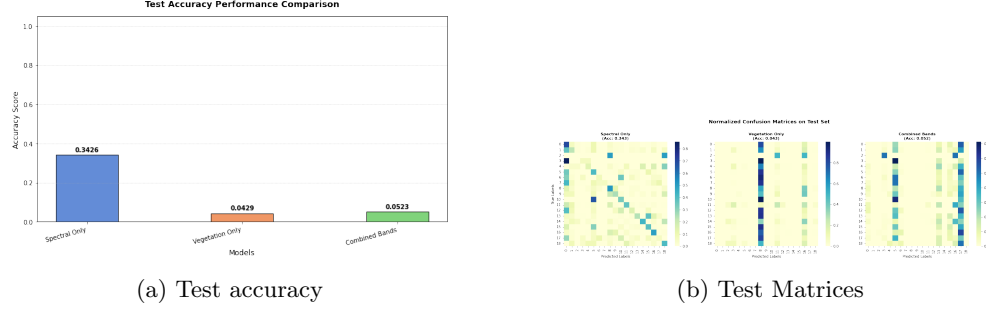
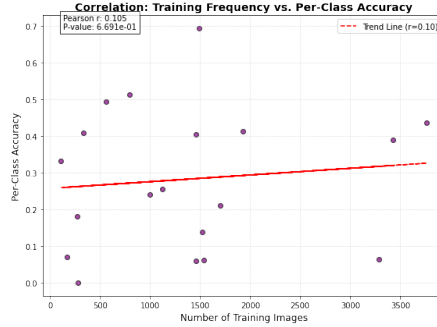


Figure 9: 5x5 — 8 months — Band combinations — test

The spectral-only model yielded the most consistent and robust learning trajectory, establishing it as the top-performing configuration ( $accuracy \approx 0.34$ ) (fig. 9a). This result serves as a critical benchmark for the PseTae architecture, providing clear guidance for future experimental iterations. These findings suggest that the model's predictive precision can be further optimized by scaling the input context, for example, by expanding temporal observation windows or changing spatial patch sizes, in addition to removing the EVI index. Such adjustments would enable a more comprehensive analysis, potentially enhancing the architecture's capacity to discriminate among complex land-cover classes.



(a) Correlation

Figure 10: Class values

Finally, to mitigate the inherent class imbalance in the training data, a cost-sensitive learning was implemented, incorporating inverse-frequency class weights into the CrossEntropyLoss function, which forces the model to prioritize minority classes during optimization. The efficacy of this balancing strategy was subsequently validated through a correlation analysis comparing training sample frequency to per-class accuracy. The resulting scatter plot, which shows a low Pearson correlation coefficient ( $r \approx 0.105$ ), confirms that there is no statistically significant relationship between sample size and model performance (fig. 10a).